

Representation and Modeling of 3D Objects - Examples

Examples: Mass – spring model

- ❑ Mass Spring Damper Cloth Simulation 1 - YouTube,
<https://www.youtube.com/watch?v=ib1vmRDs8Vw>
- ❑ Mass Spring Damper Cloth Simulation 2,
<https://www.youtube.com/watch?v=qOvb3WLAX0E>
- ❑ Fast Simulation of Mass-Spring Systems Cloth Implementation,
<https://vimeo.com/252071746>
- ❑ Simulating Mass Spring Systems,
<http://cs.ucsb.edu/~yunteng.cs/springs.html>
- ❑ Wobbly Picture using Mass-Spring system with Box2D,
<https://www.youtube.com/watch?v=3cT6ZM3KkqY>
- ❑ Top 5 Realistic Fluid Simulations,
https://www.youtube.com/watch?v=AT_TIm_Q3Qk
- ❑ Advanced Molecular & Particle Physics Simulations,
<https://www.youtube.com/watch?v=x8Fo2sIT2WA>

Examples: L-Systems

- ❑ Realtime L-System Trees 02
https://www.youtube.com/watch?v=HC8skymgu_0
- ❑ Procedurally Generated Apple Tree
<https://www.youtube.com/watch?v=ZVyi4ykVHJM>
- ❑ Fractal Trees
<https://www.youtube.com/watch?v=t65lq818c78&list=PLE4y6Wd4RpZfon6KTy4TX65BznH9MpN7w>
- ❑ Procedurally Generated Plant
https://www.youtube.com/watch?v=0_bum83FuRk&index=2&list=PLE4y6Wd4RpZfon6KTy4TX65BznH9MpN7w
- ❑ 3D Real-Time Plants Modeling with L-systems,
<https://www.youtube.com/watch?v=QJ6r8ZpQ7P8>

Examples

- ❑ Free-flowing Granular Materials with Two-way Solid Coupling ([video](#))
 - Continuum-based model
 - Efficient simulation of granular materials
 - Fully solves the internal pressure and frictional stresses in a granular material
 - Visually noticeable behaviors of granular materials
 - Freely dispersing splashes without cohesion, and a global coupling between friction and pressure
 - The full treatment of internal forces in the material also enables two-way interaction with solid bodies
 - Very small fraction of computational costs of the comparable particle-based models for granular flows

Examples

- ❑ RealFlow 2012 water simulation ([video](#))
- ❑ Stitch Meshes for Modeling Knitted Clothing with Yarn-level Detail - SIGGRAPH 2012 ([video](#))
- ❑ 3D Hair on Siggraph 2012 ([video](#))
- ❑ Realflow Flooded Corridor ([video](#))

- Use standard fluids and emitters
- Took 4 hours to simulate and mesh, 1 hour to render
- Use an opaque material for the liquid
- Expensive rendering of refractions

- ❑ Virtual Clothing

http://www0.cs.ucl.ac.uk/research/vr/Projects/3DCentre/cloth_simulation_links.htm